SUPPLEMENTAL MATERIAL

Table S1. Detailed demographics and baseline clinical characteristics for extracted studies; total and by HIV status.

Study	Data Extracted	N Total	N HIV+	N HIV-	% Male	% Male HIV+	% Male HIV-	% White	% White HIV+	% White HIV-	% Black	% Black HIV+	% Black HIV-	Avg Age	Avg Age HIV+	Avg Age HIV-	FRS HIV+	FRS HIV-
Pereira B. 2020[50]	CAC>0, CAC>100	739	739	0	92.8	92.8	-	84.2	84.2	-	7.5	7.5	-	56	56	-	-	-
Krishnam M. 2020[39]	CAC>0	143	143	0	83.2	83.2	-	72.7	72.7	-	7	7	-	46.4	46.4	-	-	-
Chandra D. 2019[29]	CAC>0; median CAC (Weston score)	234	177	57	82.1	76.3	100	-	-	-	-	-	-	49.6	48.8	52.2		
Senoner T. 2019[53]	Mean CAC, Any plaque, Plaque Presence	138	69	69	71	73.9	68.1	-	-	-	-	-	-	54	55	52	-	-
Tarr PE. 2018[l,14,57]	Median CAC, CAC>0, CAC>100, CAC progression, Plaque Presence	704	428	276	83	86	79	89	91	88	7	7	6	54	52	56	9	10
Korada SK 2017[@,4,10,38,46,47,52]	Median CAC, CAC >0, CAC >100, CAC Progression, Plaque Presence	976	602		100	100	100	58	-	-	31	-	-	54	-	-	-	-
Besutti G. 2016[#,9,15, 16,28]	Mean CAC, Median CAC, CAC >0, CAC >100, OR CAC Progression, CAC Progression,	1446	1446	0	71	71	-	-	-	-	-	-	-	48	48	-	4	-
Fitch, KV. 2016[36]	Median CAC, CAC>0, CAC>100, Plaque Presence	225	155	70	60	61	59	49	49	49	-	-	-	47	47	46	9	8
Nadel, J. 2016[49]	Median CAC	97	32	65	100	100	100	81	78	86	-	-	-	60	60	60	13	13
Chow, D. 2015[$,30,54]	Mean CAC, Median CAC, CAC>0, CAC >100, OR CAC	2833	100	2733	100	100	100	40	56	39	26	5	26	59	54	60	16	20
Abd-Elmoniem, KZ. 2014[26]	Plaque Presence	46	35	11	48	54	27	37	34	45	46	49	36	23	22	25	-	-
Longenecker, CT. 2014[44]	CAC>0	147	147	0	78	78	-	-	-	-	69	69	-	46	46	-	3	-
Baker, JV. 2014[13]	CAC>0, CAC progression	436	436	0	78	78	-	59	59	-	27	27	-	42	42	-	5	-
Kristoffersen US. 2013[40]	Mean CAC, CAC >0, CAC >100	210	105	105	89	89	89	-	-	-	-	-	-	47	47	47	9	8
Lai S. 2013[^,12,41-43]	Mean CAC, CAC >0, CAC progression, Plaque Presence	848	848	0	63	63	-	0	0	-	100	100	-	46	46	-	4	-
Pereyra, F. 2013[51]	Median CAC, CAC>0, Plaque Presence	152	103	49	68	69	67	-	-	-	-	-	-	49	49	48	10	9
Hsue, PY. 2012[37]	CAC >0, CAC >100	311	253	58	88	89	83	62	62	62	22	22	21	49	49	48	4	3
Duarte, H. 2012[33]	Mean CAC	52	26	26	85	85	85	48	42	54	40	54	27	53	53	52	12	10
Fitch, K. 2012[35]	Mean CAC	50	46	0	76	76	0	64	64	-	32	32	-	47	47	-	-	-
d'Ettorre G. 2012[32]	Mean CAC, CAC>100	55	55	0	86	86	-	-	-	-	-	-	-	48	48	-	4	-
Subramanian S. 2012[55]	Median CAC, CAC>0	54	27	27	93	93	93	-	-	-	-	-	-	53	52	54	-	-
Falcone EL. 2011[&,11,34,45,55]	Mean CAC, Median CAC, CAC >0, CAC >100, CAC Progression,	334	334	0	74	74	-	53	53	-	34	34	-	44	44	47	7	-
Crum-Cianflone N. 2011[31]	CAC>0, CAC>100	223	223	0	96	96	-	49	49	-	23	23	-	43	43	-	-	-
Monteiro VS. 2011[48]	CAC>0	53	53	0	51	51	-	-	-	-	-	-	-	43	43	-	-	-
Vilela FD. 2011[58]	Median CAC, CAC>0	40	40	0	53	53	-	-	-	-	-	-	-	46	46	-	4	-
Acevedo M. 2002[27]	Median CAC, CAC>0	85	17	68	-	-	-	-	-	-	-	-	-	46	46	46	10	-
Talwani R. 2002[56]	CAC>0	240	60	180	100	100	100	65	65	65	27	27	27	47	47	47	-	-
Pooled / Combined*	**All**	**10867**	**6699**	**4168**	**86**	**79**	**96**	**51**	**53**	**48**	**32**	**37**	**24**	**52**	**49**	**57**	**6**	**18**

Study	% on ART	Avg ART Dur (yrs)	% AIDS (current)	% AIDS (Ever)	CD4 Count	% DM Total	% DM HIV+	% DM HIV-	% HTN Total	% HTN HIV+	% HTN HIV-	% Smoking Total	% Smoking HIV+	% Smoking HIV-	Avg SBP HIV+	Avg SBP HIV-	Excluded Prior CAD
Pereira B. 2020[50]	-	14	-	-	633	5.8	5.8	-	21.4	21.4	-	42.9	42.9	-	130	-	Yes
Krishnam M. 2020[38]	65	-	-	-	-	16.1	16.1	-	13.3	13.3	-	32.2	32.2	-	122.9	-	Yes
Chandra D. 2019[29]	87	-	-	-	602	4.7	2.8	7	-	-	-	43.2	24.9	100	126	132.1	No
Senoner T. 2019[53]	-	-	-	-	669	9	10	9	37	35	39	80	81	80	-	-	Yes
Tarr PE. 2018[l,,14,57]	93	10.1	-	21	598	7	6	9	43	34	56	28	35	17	-	-	Yes
Korada SK 2017[@,4,10,38,46,47,52]	56	6.6	-	14	519	8	8	8	19	15	25	63	64	61	126	130	No
Besutti G. 2016[#,,9,15,16,28]	100	0.91	-	-	571	13	13	-	38	38	-	39	39	-	-	-	No
Fitch, KV. 2016[36]	99	-	-	-	552	9	10	7	22	24	16	43	44	40	119	117	Yes
Nadel, J. 2016[49]	-	19	-	47	649	24	28	22	64	66	63	18	25	14	128	130	No
Chow, D. 2015[$,,30,54]	-	-	-	-	498	14	9	14	-	-	-	16	24	16	126	125	Yes
Abd-Elmoniem, KZ. 2014[26]	71	15	14	-	502	-	-	-	-	-	-	-	-	-	123	116	Yes
Longenecker, CT. 2014[44]	-	5.3	-	-	613	-	-	-	-	-	-	63	63	-	121	-	Yes
Baker, JV. 2014[13]	78	2.8	-	-	481	9	9	-	-	-	-	41	41	-	-	-	No
Kristoffersen US. 2013[40]	100	8.9	-	-	636	-	-	-	-	-	-	37	37	37	131	123	Yes
Lai S. 2013 [^,,12,41-43]	-	3	-	-	368	4	4	-	12	12	-	83	83	-	117	-	Yes
Pereyra, F. 2012[51]	100	8.5	-	-	571	9	12	4	23	26	16	40	40	41	121	117	Yes
Hsue, PY. 2012[57]	68	5.7	14	-	471	5	6	2	27	28	21	62	62	-	-	-	Yes
Duarte, H. 2012[33]	81	14	-	-	582	25	35	15	60	65	54	31	31	31	126	135	Yes
Fitch, K. 2012[35]	-	-	-	-	583	-	-	-	-	-	-	-	-	-	121	-	Yes
d'Ettorre G. 2012[32]	89	9.4	-	-	493	-	-	-	-	-	-	-	-	-	-	-	No
Subramanian S. 2012[55]	100	12.3	-	-	641	-	-	-	17	19	15	15	22	7	124	121	Yes
Falcone EL. 2011 [&,,11,34,45,55]	74	2.7	-	-	454	-	-	-	-	-	-	50	50	-	119	-	No
Crum-Cianflone N. 2011[31]	83	6.4	-	-	586	6	6	-	30	30	-	17	17	-	-	-	No
Monteiro VS. 2011[48]	100	4.9	-	-	-	4	4	-	23	23	-	19	19	-	-	-	Yes
Vilela FD. 2011[58]	100	-	-	-	-	10	10	-	55	55	-	35	35	-	123	-	Yes
Acevedo M. 2002[27]	100	3.1	-	-	442	-	18	-	-	47	-	-	53	-	-	-	Yes
Talwani R. 2002[56]	94	2.2	-	-	483	-	-	-	-	-	-	-	-	-	-	-	No
Pooled/Combined*	**91**	**6**	**14**	**18**	**543**	**10**	**10**	**13**	**32**	**30**	**42**	**36**	**45**	**20**	**124**	**125**	

Table S2. CAC and Plaque Data among HIV(+) participants in the studies (% = percent of participants; CAC assess by non-contrast CT; plaque assess by CTA).

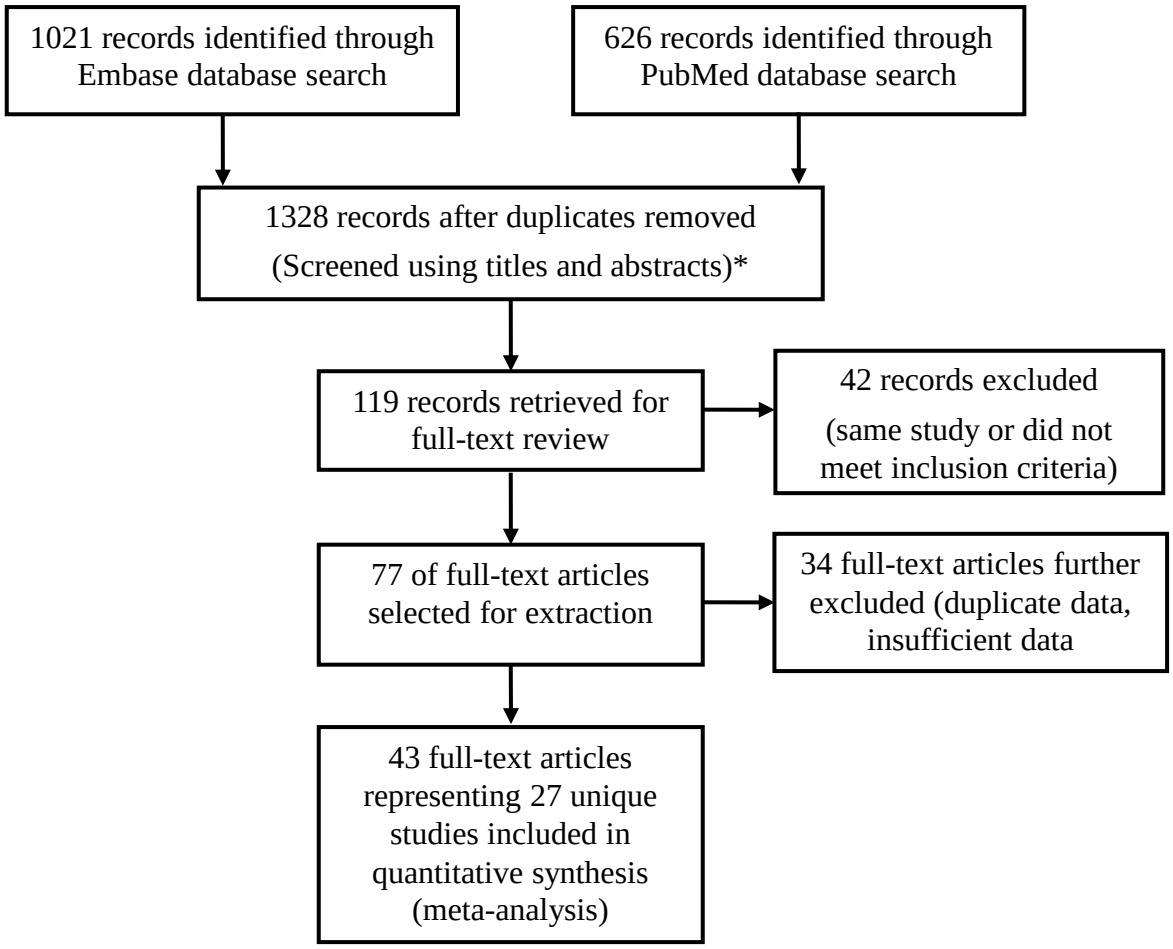
Study	Included HIV-Control?	N HIV+	Mean CAC	Median CAC	% CAC > 0	% CAC > 100	% CAC Progression	% Plaque Present	% Calcified Plaque Present	% Noncalcified Plaque Present
Pereira B. 2020[50]	No	739			53	18				
Krishnam M. 2020[39]	No	143			29					
Chandra D. 2019[29]	Yes	177			44					
Senoner T. 2019[53]	Yes	69	149.4 ± 287.1					84		65
Tarr PE. 2018 [!,57]	Yes	428		47 [14,183]	53	19		53	37	37
Tarr PE. 2020 [!,14]	Yes	340					7	24	20	21
Post WS. 2014[@,52]	Yes	450						78	35	63
Kingsley LA. 2008[@,4]	Yes	615			47				56	
Kingsley LA. 2015 [@,10]	Yes	376		0 [0,24]			21			
Korada SK. 2017 [@,38]	Yes	602				22				
Zona S. 2012[#,9]	No	240		0 [0,11]			10			
Besutti G. 2016[#,28]	No	1446			35	10				
Guaraldi G. 2011[#,15]	No	876	40 ± 155							
Guaraldi G. 2012 [#,16]	No	132		0 [0-974]			34			
Fitch KV. 2016[36]	Yes	155	23 ± 6	0 [0,9]	36	7		54	11	44
Nadel J. 2016[49]	Yes	32		56 [0,545]						
Chow D. 2015[$,30]	Yes	100	151.25 ± 390.06		58					
Shikuma CM. 2014 [$,54]	No	130		0 [0,47.66]	47	20				
Abd-Elmoniem, KZ. 2014[26]	Yes	35						20		20
Longenecker CT. 2014[44]	No	147			37					
Baker JV. 2014[13]	No	436					13			
Kristoffersen US. 2013[40]	Yes	105	54.3 ± 16.7		37	11				
Lai S. 2009[^,41]	No	176						30		
Lai S. 2005 [^,43]	Yes	124			36					
Lai S. 2013 [^,41]	No	848			28					
Lai H. 2012 [^,12]	No	119					12			
Pereyra F. 2013[51]	Yes	103		0 [0,16]	41			60		
Hsue PY. 2012[37]	Yes	253			37	16				
Duarte H. 2012[33]	Yes	26	114 ± 218							
Fitch K. 2012[35]	No	46	48 ± 76							
d'Ettorre G. 2012[32]	No	55	21.25 ± 63.53			11				
Subramanian S. 2012[55]	Yes	27		0 [0,92.6]	37					
Volpe GE. 2013[&,11]	No	211		0.4 [0,5.2]			33			
Mangili A. 2007 [&,45]	No	314				8				
Falcone EL. 2010 [&,34]	No	298	46 ± 240.7							
Falcone EL. 2011 [&,59]	No	334			52					
Crum-Cianflone N. 2011[31]	No	223			34	8				
Monteiro VS. 2011[48]	No	53			21					
Vilela FD. 2011[58]	No	40		0	33					
Acevedo M. 2002[27]	Yes	17		14.4 [1,131.5]	76					
Talwani R. 2002[56]	Yes	60			33					
All (total N, pooled)	-	**6699 (total)**	**67 (mean)**	-	**40**	**19**	**13 / 13 / 21%**	**64**	**31**	**49**

Table S3. Newcastle-Ottowa Scoring Table.

Study	Selection 1	Selection 2	Selection 3	Selection 4	Comparability a	Comparability b	Outcome / Exp -1	Outcome / Exp - 2	Outcome / Exp - 3	Total Score
Pereira B. 2020[50]	1	0	1	1	0	1	1	0	0	5
Krishnam M. 2020[39]	0	0	1	1	0	1	1	0	0	4
Chandra D. 2019[29]	1	1	1	1	1	1	1	1	1	9
Senoner T. 2019[53]	1	1	1	1	1	1	1	0	0	7
Tarr PE. 2018 [!,14,57]	1	1	1	1	1	1	1	1	1	9
Korada SK. 2017 [@,4,10,38,46,47,52]	1	1	1	1	1	1	1	0	0	7
Besutti G. 2016 [#,9,15,16,28]	1	0	1	1	0	1	1	0	0	5
Fitch, KV. 2016[36]	1	1	1	1	1	1	1	0	0	7
Nadel, J. 2016[49]	1	1	1	1	1	1	1	1	1	9
Chow, D. 2015 [$,,30,54]	1	1	1	1	1	1	1	0	0	7
Abd-Elmoniem, KZ. 2014[26]	1	1	1	1	1	1	1	0	0	7
Longenecker, CT. 2014[44]	1	0	1	1	0	1	1	0	0	5
Baker, JV. 2014[13]	1	0	1	1	0	1	1	1	1	7
Kristoffersen US. 2013[40]	1	1	1	1	1	1	1	0	0	7
Lai S. 2013 [^,12,41-43]	1	0	1	1	0	1	1	0	0	5
Pereyra, F. 2013[51]	1	1	1	1	1	1	1	0	0	7
Hsue, PY. 2012[37]	1	1	1	1	1	1	1	0	0	7
Duarte, H. 2012[33]	1	1	1	1	1	1	1	0	0	7
Fitch, K. 2012[35]	1	0	1	1	0	1	1	1	1	7
d'Ettorre G. 2012[32]	1	0	1	1	0	1	1	0	0	5
Subramanian S. 2012[55]	1	1	1	1	1	1	1	0	0	7
Falcone EL. 2011 [&,11,34,45,55]	1	0	1	1	0	1	1	0	0	5
Crum-Cianflone N. 2011[31]	1	0	1	1	0	1	1	0	0	5
Monteiro VS. 2011[48]	1	0	1	1	0	1	1	0	0	5
Vilela FD. 2011[58]	1	0	1	1	0	1	1	0	0	5
Acevedo M. 2002[27]	1	1	1	1	1	1	1	0	0	7
Talwani R. 2002[56]	1	1	1	1	1	1	1	0	0	7

*Represents sub-total for N and weighted average for % (weighted by N)
! - Tarr PE. 2018 contains the same study population as Tarr PE. 2020
@ - Korada 2017 contains same study population as Post WS 2014, Monroe AK 2012, Metkus TS 2015, Kingsley 2015, Kingsley LA 2008
- Besutti G 2016 contains same study population as Guaraldi G 2011, Guaraldi G 2012, Zona S 2012)
$ - Chow C D 2015 contains same study population as Shikuma C 2014
^ - Lai S 2013 contains same study population as Lai S 2009, Lai S 2005, Lai H 2012; (Dates of subject enrollment and analysis did not overlap for Lai 2005 and Lai 2013)
& - Falcone EL 2011 contains same study population as Volpe GE 2013, Mangili A 2007, Falcone EL 2010

Figure S1. Study Selection Process.



*No additional records identified through other sources

Supplement Figure 2

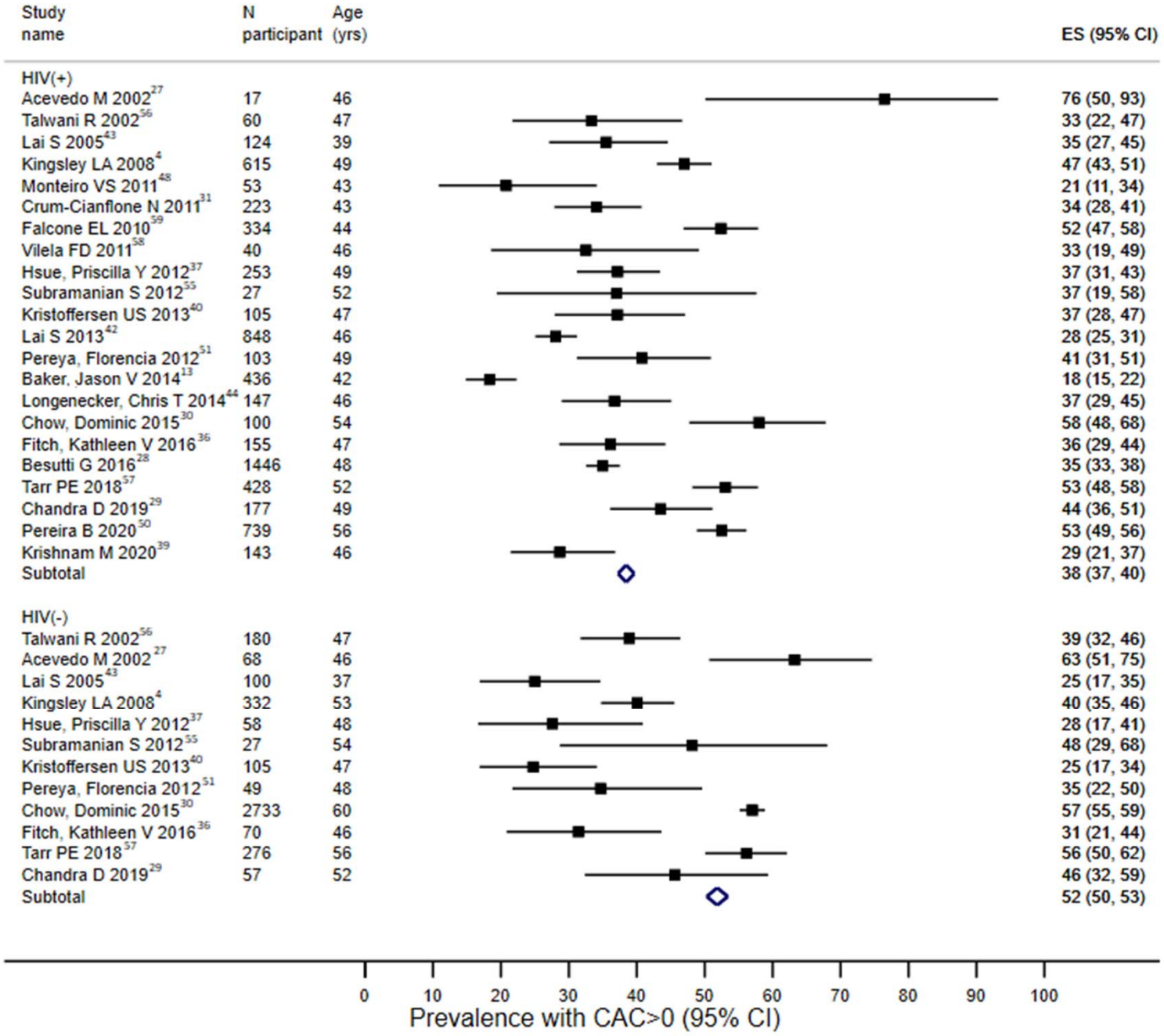


Figure S3. Prevalence of coronary calcium >100 by HIV status (studies with HIV negative controls).

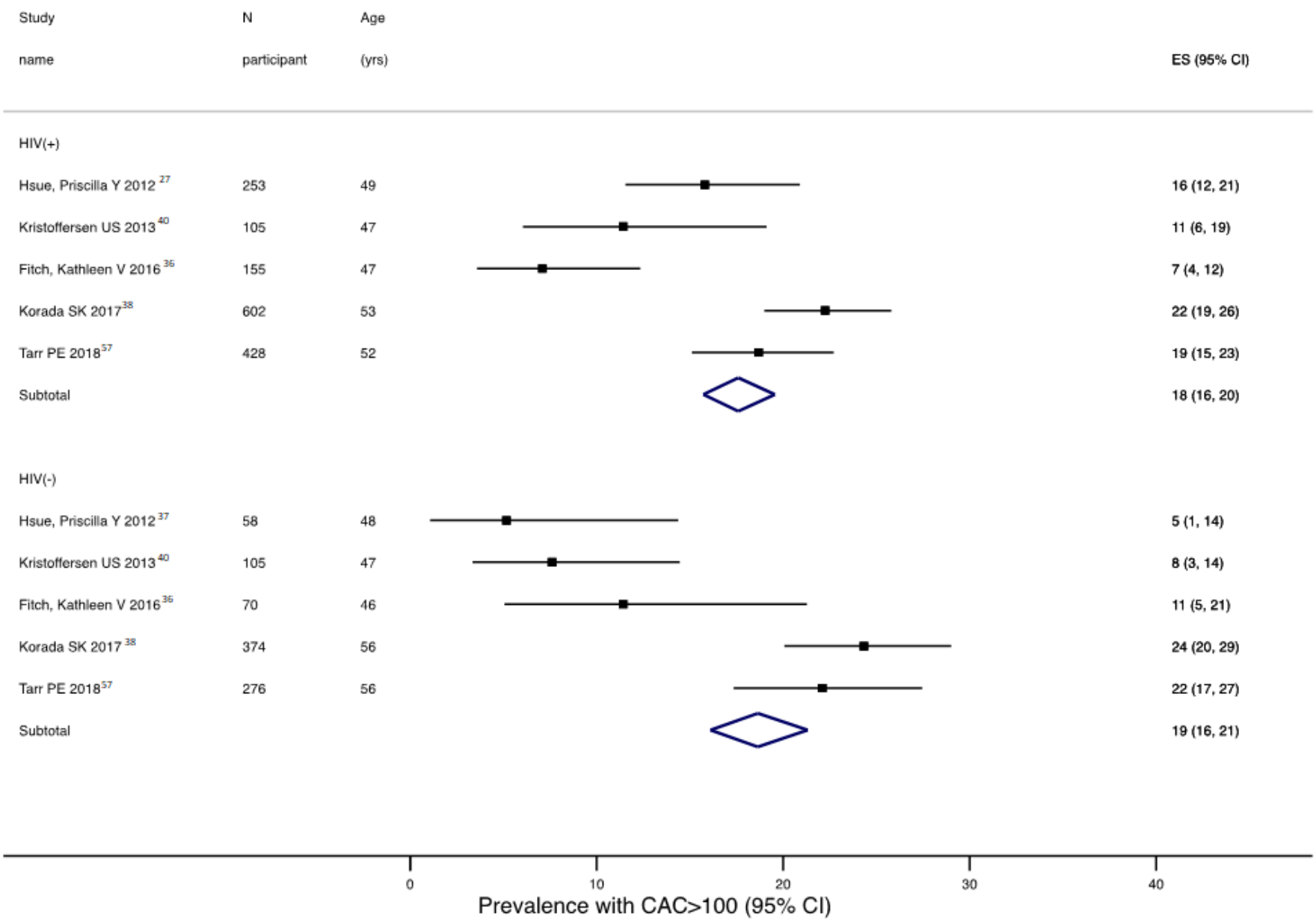


Figure S4. CAC progression among HIV positive individuals.

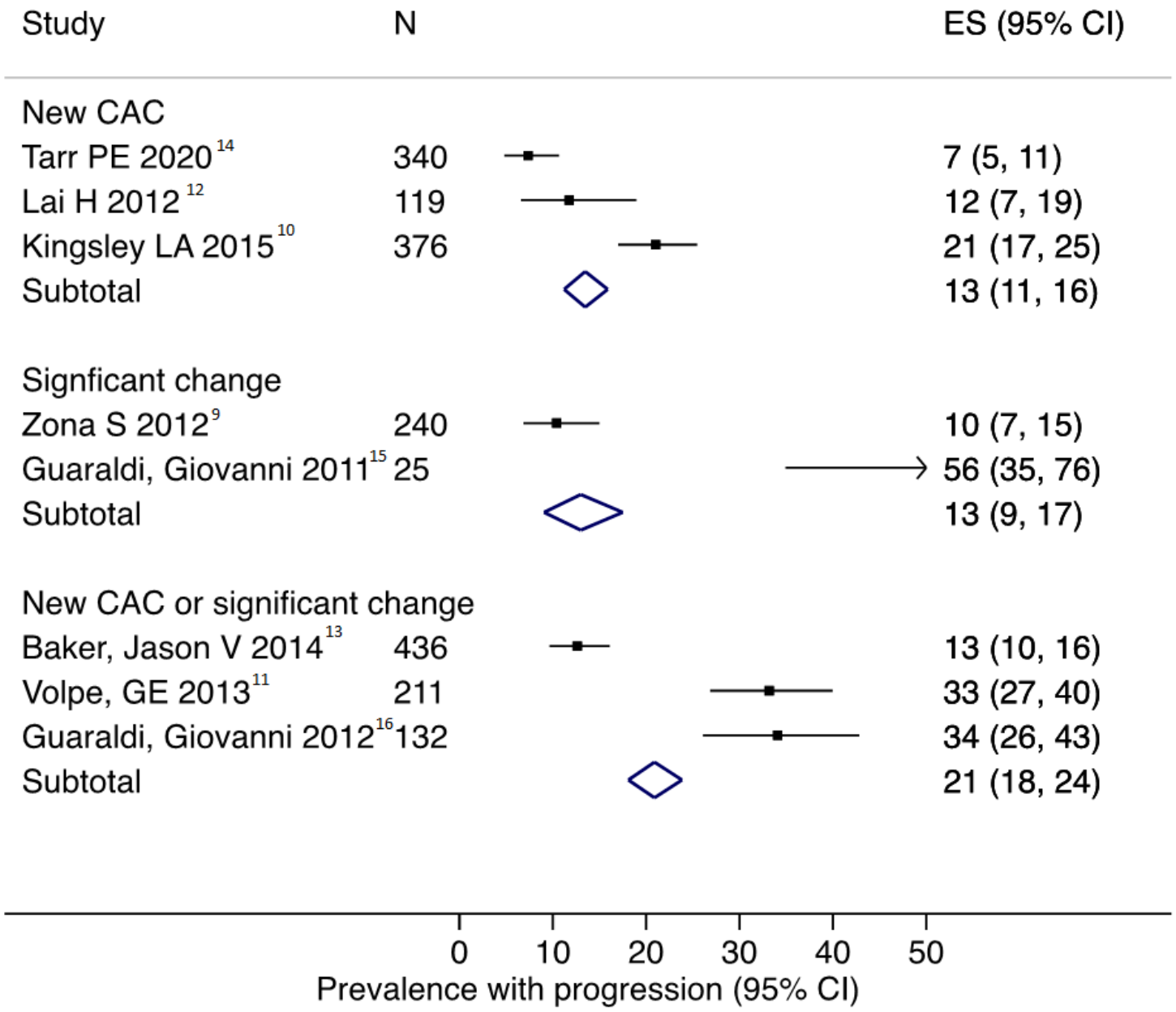
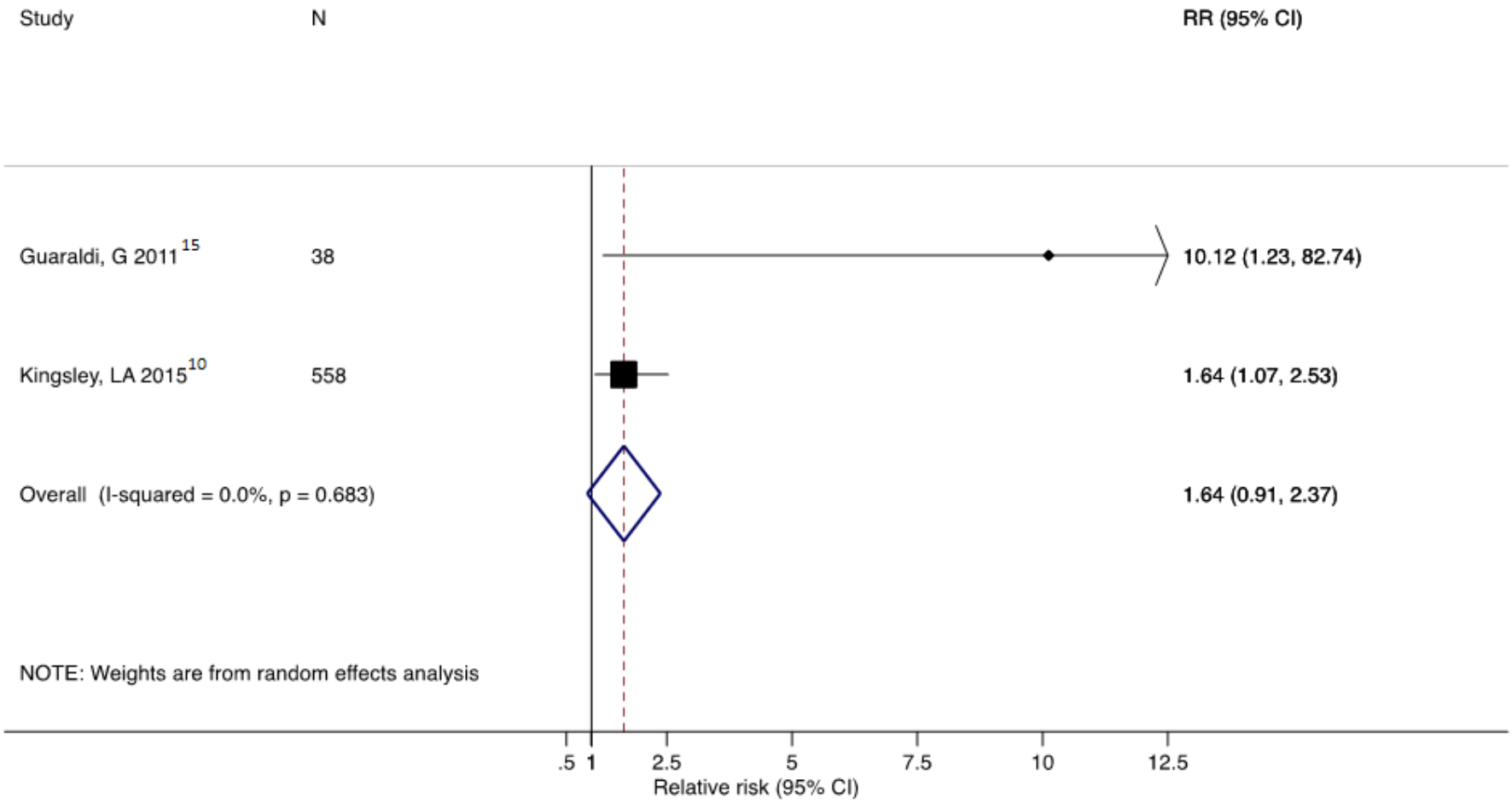


Figure S5. Odds ratio of plaque progression comparing HIV positive vs. HIV negative participants.



Supplement Figure 6

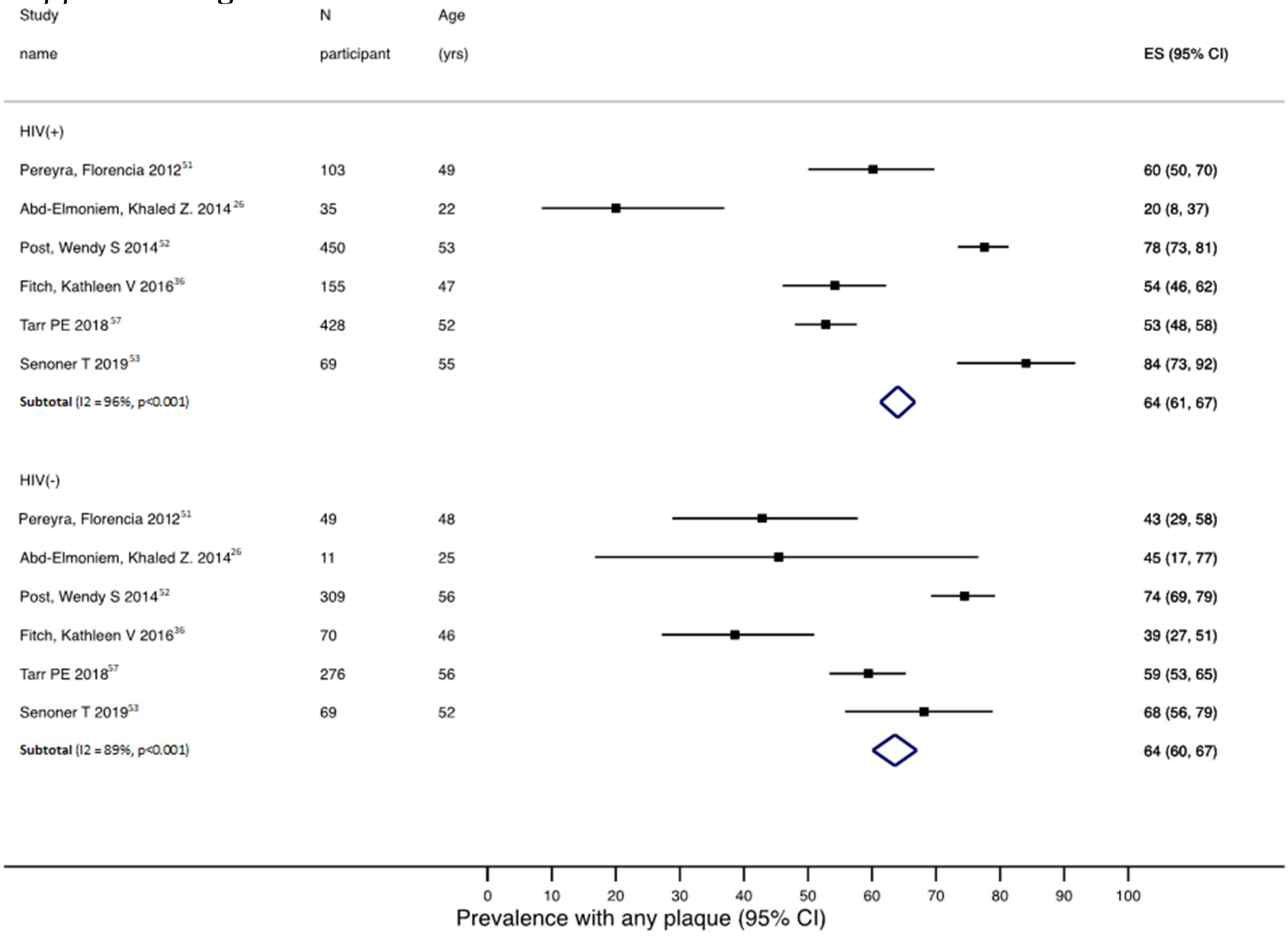


Figure S7. **Meta-regression of coronary calcium presence study estimates by additional study-level characteristics.**

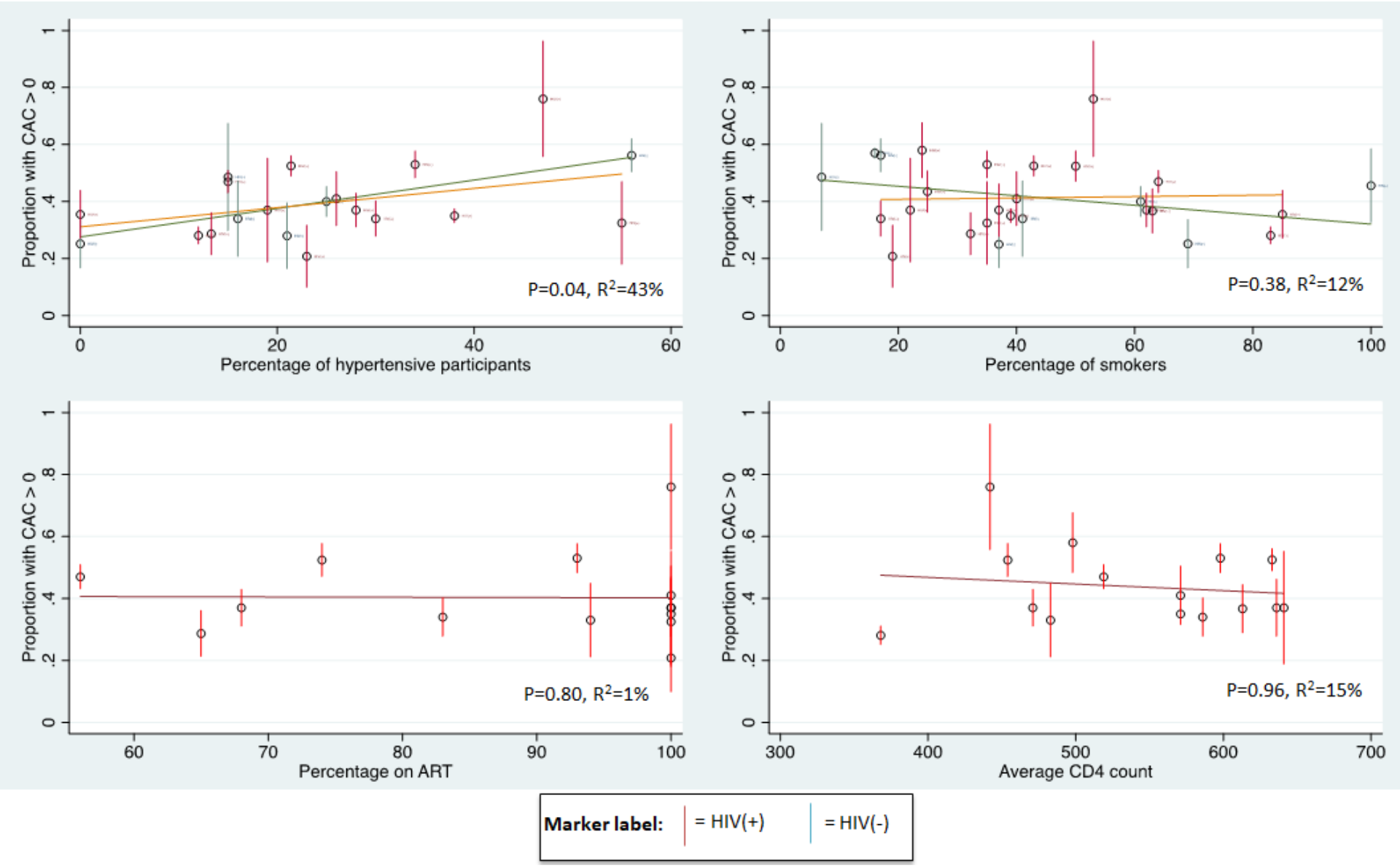


Figure S8. Subgroup analyses comparing HIV(+) vs. HIV(-) individuals, restricting to studies with average value less than the median for each group

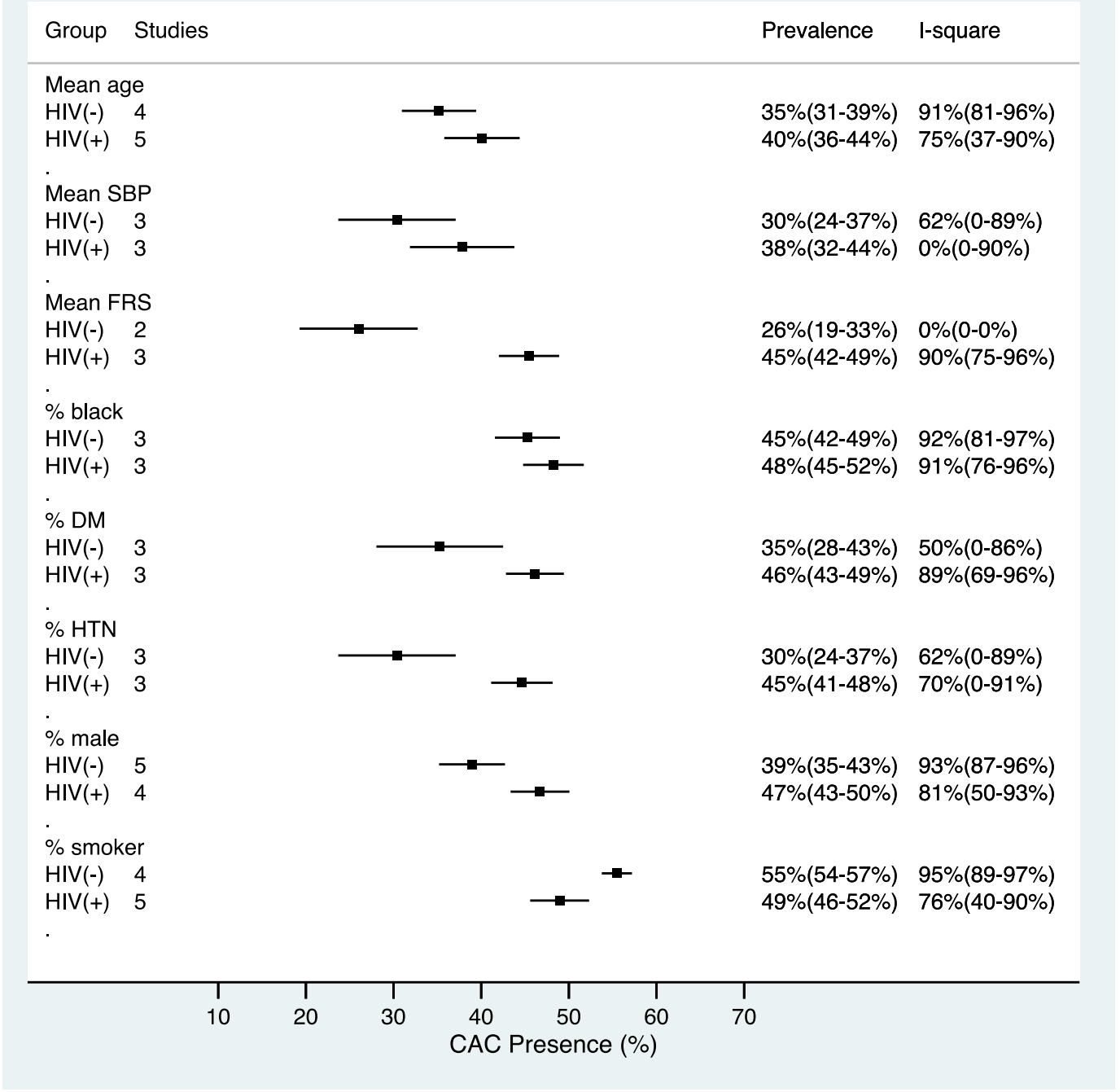


Figure S9. Subgroup analyses comparing HIV(+) vs. HIV(-) individuals, restricting to studies with average value more than the median for each group

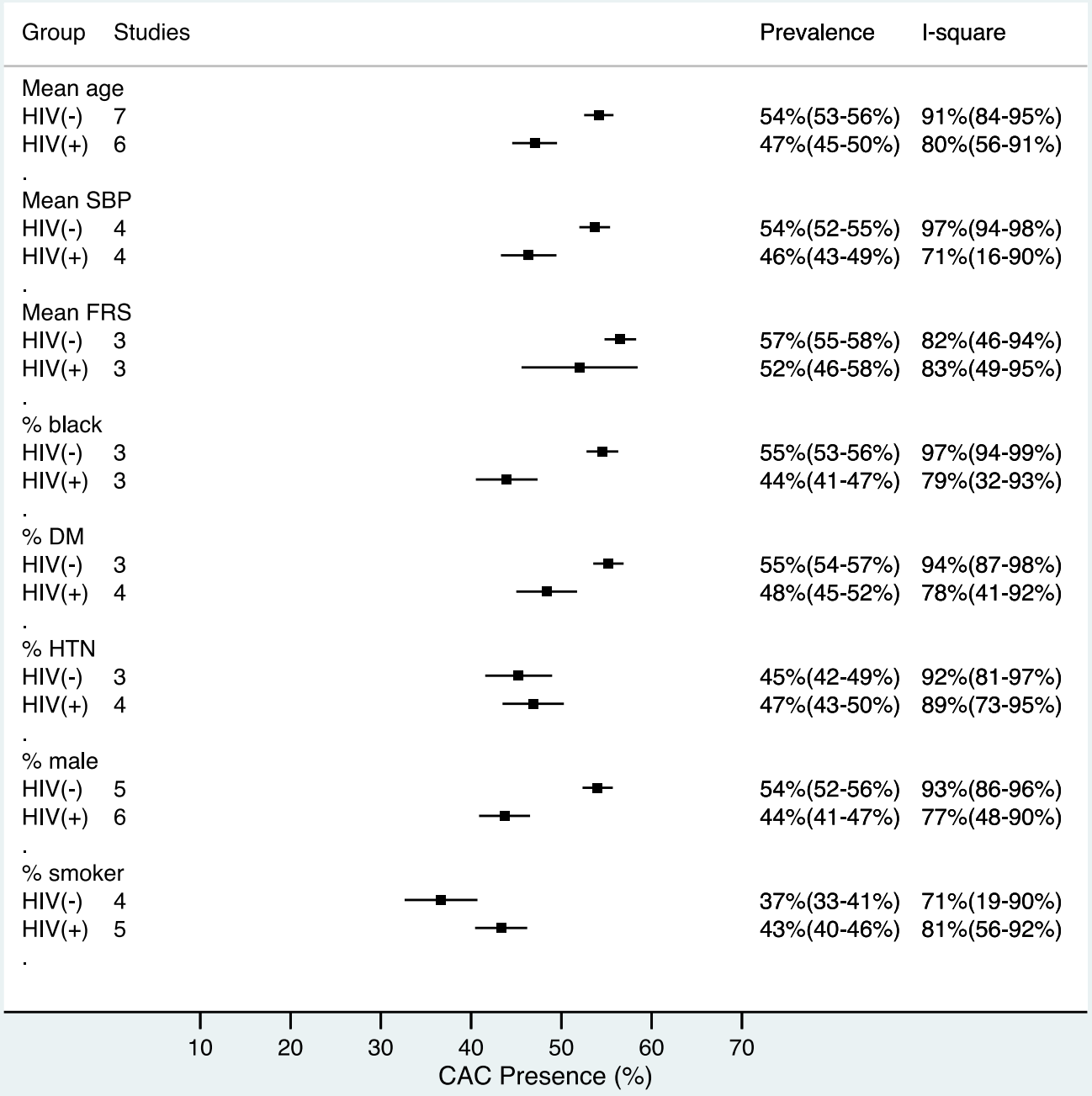


Figure S10. Funnel plot of studies included in meta-analyses by outcome.

